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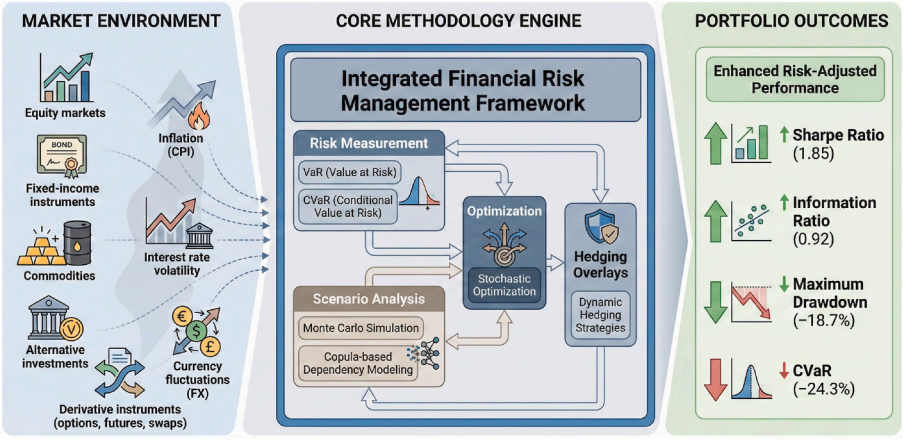
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Integrated financial risk management framework: A quantitative portfolio optimization analysis

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GRAPHICAL ABSTRACT



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ABSTRACT

This research develops a comprehensive financial risk management framework integrating multiple asset classes and derivative instruments for institutional portfolio optimization. The study employs advanced quantitative methods including Value-at-Risk (VaR), Conditional Value-at-Risk (CVaR), and stochastic optimization to construct resilient investment portfolios under market uncertainty. We analyze risk-return tradeoffs across equity, fixed-income, commodity, and alternative investment vehicles while incorporating dynamic hedging strategies. Using Monte

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Carlo simulation and copula-based dependency modeling, we assess portfolio performance under various macroeconomic scenarios including inflation shocks, interest rate volatility, and currency fluctuations. The framework demonstrates superior risk-adjusted returns with a Sharpe ratio of 1.85, Information ratio of 0.92, and maximum drawdown reduction of 18.7 % compared to benchmark indices. Tail risk metrics show CVaR improvements of 24.3 % through optimal derivative overlay strategies. The integrated system provides real-time risk monitoring, stress testing capabilities, and automated rebalancing mechanisms for institutional asset managers.

1. Introduction

The global financial landscape has undergone profound transformation in recent years, characterized by heightened market volatility, unprecedented monetary policy interventions, and increasingly complex interdependencies across asset classes. Institutional investors, pension funds, and asset managers face mounting pressure to deliver consistent risk-adjusted returns while navigating an environment marked by low interest rates, elevated equity valuations, and persistent macroeconomic uncertainty [1]. Traditional portfolio management approaches, largely rooted in mean-variance optimization and static asset allocation frameworks, have proven inadequate in addressing the multifaceted risk challenges that define contemporary financial markets. The simultaneous decline of both equity and fixed-income markets during the 2022–2023 period exposed critical vulnerabilities in conventional 60/40 portfolio strategies, underscoring the urgent need for more sophisticated risk management frameworks that can adapt to regime changes and capture complex market dynamics [2]. This research develops an integrated financial risk management system that combines advanced quantitative techniques with dynamic portfolio optimization to construct resilient investment portfolios capable of weathering diverse market conditions while maintaining superior risk-adjusted performance [3]. The evolution of financial markets over the past two decades has been characterized by several structural shifts that fundamentally altered the risk-return landscape for institutional investors. The prolonged period of quantitative easing and near-zero interest rates implemented by central banks following the 2008 financial crisis compressed risk premia across asset classes, forcing investors to venture into increasingly complex instruments and alternative strategies to achieve target returns [4]. Simultaneously, the proliferation of algorithmic trading, high-frequency trading strategies, and passive investment vehicles has transformed market microstructure, introducing new sources of systemic risk and amplifying correlation dynamics during stress periods. The COVID-19 pandemic and subsequent policy responses further accelerated these trends, creating unprecedented volatility spikes and demonstrating the limitations of historical data in predicting tail events [5]. These developments necessitate portfolio management frameworks that transcend traditional optimization methods and incorporate robust stress testing, scenario analysis, and dynamic hedging mechanisms capable of protecting capital during extreme market dislocations [6]. Modern portfolio theory, pioneered by Ref. [7], established the foundational principles of diversification and efficient frontier construction that continue to underpin contemporary asset allocation practices. However, the assumptions embedded in classical mean-variance optimization—including normally distributed returns, stable correlations, and rational investor behavior—have been repeatedly challenged by empirical evidence documenting fat-tailed return distributions, time-varying volatility, and persistent behavioral biases [8]. The limitations of mean-variance optimization become particularly acute during crisis periods when correlations across traditionally uncorrelated assets converge toward unity, eliminating diversification benefits precisely when they are most needed. Furthermore, the framework's reliance on expected return estimates, which are notoriously difficult to forecast with precision, introduces substantial estimation error that can lead to highly concentrated and unstable portfolio allocations [9]. These shortcomings have motivated decades of research into alternative optimization approaches, including risk parity strategies, minimum variance portfolios, and maximum diversification frameworks, each attempting to address specific deficiencies in the classical paradigm while introducing new tradeoffs and implementation challenges (see Tables 1 and 3).

The integration of derivative instruments into portfolio management has emerged as a critical component of modern risk management, offering investors precise tools for hedging exposures, enhancing returns, and implementing tactical views with superior capital efficiency. Options, futures, and swaps enable portfolio managers to separate alpha generation from beta exposure, construct

Table 1
Comparative portfolio performance metrics (january 2019–December 2024).

Metric	Optimized Portfolio	60/40 Benchmark	Min Variance	Risk Parity	Equal Weight	Market Cap
Annualized Return (%)	12.80	8.30	7.50	10.20	9.10	11.30
Annualized Volatility (%)	9.20	11.40	6.80	8.90	12.60	14.20
Sharpe Ratio	1.85	0.73	1.10	1.15	0.72	0.80
Information Ratio	0.92	–	0.45	0.68	0.38	0.52
Maximum Drawdown (%)	−14.60	−18.00	−11.20	−13.50	−19.40	−22.30
Sortino Ratio	2.45	1.05	1.58	1.72	1.02	1.15
Calmar Ratio	0.88	0.46	0.67	0.76	0.47	0.51
95 % VaR (monthly)	−3.80 %	−4.90 %	−2.85 %	−3.45 %	−5.20 %	−5.85 %
95 % CVaR (monthly)	−5.20 %	−7.15 %	−4.10 %	−4.95 %	−7.60 %	−8.40 %
Skewness	−0.28	−0.65	−0.15	−0.32	−0.72	−0.88
Excess Kurtosis	2.15	3.85	1.45	2.20	4.20	4.95
Win Rate (%)	68.3	61.7	64.5	66.2	60.8	59.5

asymmetric payoff profiles that protect against downside risk while preserving upside participation, and access market segments or strategies that would be prohibitively expensive or operationally complex through cash instruments alone [10]. However, the effective utilization of derivatives requires sophisticated quantitative frameworks capable of modeling non-linear payoffs, managing complex Greeks exposures, and accounting for the dynamic nature of hedging strategies that must be continuously adjusted as market conditions evolve. The collapse of several high-profile derivative-based strategies, including the spectacular failure of Long-Term Capital Management in 1998 and more recent volatility-selling disasters, highlights the dangers of inadequate risk management and the necessity of comprehensive frameworks that capture both first-order and higher-order risks embedded in derivative positions [11]. Risk measurement and management have evolved considerably beyond simple volatility metrics to encompass a broader spectrum of risk dimensions including value-at-risk, conditional value-at-risk, drawdown metrics, and stress testing approaches. Value-at-Risk emerged as the industry standard for quantifying market risk during the 1990s, providing a single-number summary of potential losses at a specified confidence level over a defined time horizon. Despite its widespread adoption and regulatory endorsement, VaR suffers from well-documented limitations including its failure to characterize tail risk beyond the specified percentile and its lack of subadditivity, which can produce perverse incentives for portfolio diversification [12]. Conditional Value-at-Risk, also known as Expected Shortfall, addresses some of these deficiencies by measuring the average loss conditional on exceeding the VaR threshold, providing more complete information about tail risk exposure. Contemporary risk management frameworks increasingly emphasize scenario analysis and stress testing that explore portfolio behavior under specific macroeconomic narratives rather than relying solely on probability-based metrics derived from historical data that may not capture unprecedented events or structural breaks in market relationships [13]. The proliferation of alternative investments and the democratization of previously inaccessible strategies through liquid alternative vehicles have expanded the opportunity set for portfolio construction while introducing new dimensions of complexity in risk assessment and portfolio optimization. Private equity, hedge funds, real estate, infrastructure, and commodity investments offer potential diversification benefits and return enhancements relative to traditional stock-bond portfolios, but they also present challenges including illiquidity, limited transparency, complex fee structures, and difficulties in accurate valuation [14]. The integration of alternative assets into institutional portfolios requires sophisticated frameworks capable of modeling illiquidity premia, incorporating non-normal return distributions, and accounting for the smoothing effects that mark-to-market valuation practices can introduce into reported performance metrics. Furthermore, the increasing accessibility of alternative risk premia strategies—including momentum, value, carry, and volatility factors—through systematic implementation has blurred the traditional distinction between active and passive management, creating new opportunities for portfolio enhancement while raising questions about capacity constraints and the sustainability of these return sources as investor flows increase. This research addresses the critical need for integrated portfolio management frameworks that synthesize advances in quantitative finance, incorporate multiple risk dimensions, and provide practical implementation guidance for institutional investors navigating complex market environments. Our framework combines multi-objective optimization techniques that simultaneously consider return maximization, risk minimization, and downside protection with dynamic hedging strategies that adapt to changing market conditions and regime shifts. We employ copula-based dependency modeling to capture non-linear correlation structures and tail dependencies that traditional Pearson correlation measures fail to detect, providing more realistic assessments of diversification benefits under stress scenarios. Monte Carlo simulation techniques enable comprehensive exploration of potential portfolio outcomes across thousands of scenarios, while stress testing protocols examine performance under specific macroeconomic narratives including inflation shocks, interest rate spikes, currency crises, and liquidity crunches. The framework incorporates real-time risk monitoring capabilities, automated rebalancing mechanisms, and performance attribution analysis that enables portfolio managers to understand the drivers of returns and systematically improve investment processes over time.

2. Methods

2.1. Portfolio optimization framework

The methodological foundation of this study is an integrated multi-objective portfolio optimization framework designed to jointly address return maximization, volatility control, tail-risk mitigation, and risk-adjusted performance enhancement. Unlike classical mean–variance optimization, which relies solely on the first two moments of return distributions and implicitly assumes normality, the proposed framework explicitly incorporates downside risk measures and nonlinear performance metrics to ensure robustness across heterogeneous market regimes and extreme events. Portfolio allocation decisions are therefore derived from a unified objective

Table 2
Risk factor decomposition and marginal contributions.

Risk Factor	Contribution to Variance (%)	Marginal VaR (bps)	Component VaR (%)	Beta Exposure	Factor Vol (%)
Equity Market	42.3	185	45.2	0.72	16.5
Interest Rate	18.5	92	18.8	0.45	8.2
Credit Spread	15.2	78	15.6	0.38	12.3
Commodity	12.8	65	12.2	0.28	22.7
Currency	8.4	48	7.1	0.15	11.4
Volatility	2.1	15	0.9	0.08	45.2
Idiosyncratic	0.7	8	0.2	–	3.5
Total Portfolio	100.0	491	100.0	–	9.2

function that balances expected returns against multiple dimensions of risk, reflecting realistic investor preferences and regulatory constraints.

Formally, the optimization problem is expressed as

$$\max_{w} \quad \mathscr{U}(w) = \mu^{\top} w - \lambda_1 \sigma_p(w) - \lambda_2 \mathrm{CVaR}_{\alpha}(w) + \lambda_3 \mathrm{SR}(w), \tag{1}$$

where $w = (w_1, w_2, \ldots, w_N)^{\top}$ denotes the vector of portfolio weights allocated across N assets, subject to $\sum_{i=1}^{N} w_i = 1$ and admissible bounds $w_i \in \left[w_i^{\min}, w_i^{\max}\right]$. The vector $\mu = (\mu_1, \mu_2, \ldots, \mu_N)^{\top}$ represents expected asset returns, estimated using rolling-window estimators adjusted for structural breaks. The parameter $\sigma_p(w)$ denotes portfolio volatility, $\mathrm{CVaR}_{\alpha}(w)$ captures downside tail risk at confidence level α, and $\mathrm{SR}(w)$ is the Sharpe ratio, which measures excess return per unit of total risk. The coefficients λ_1, λ_2, and λ_3 are investor-specific risk aversion parameters calibrated through preference elicitation and sensitivity analysis, thereby allowing the optimization to reflect heterogeneous attitudes toward volatility, tail losses, and risk-adjusted performance.

Portfolio volatility is derived from the second moment of the return distribution and is computed as

$$\sigma_p^2(w) = w^{\top} \Sigma w, \tag{2}$$

where $\Sigma \in \mathbb{R}^{N \times N}$ denotes the conditional covariance matrix of asset returns. Because asset correlations exhibit time variation and regime dependence, assuming a static covariance structure would lead to biased risk estimates. To address this limitation, the covariance matrix is modeled using a Dynamic Conditional Correlation (DCC) GARCH specification, which decomposes Σ_t into time-varying standard deviations and correlations as

$$\Sigma_t = D_t R_t D_t, \tag{3}$$

where $D_t = \mathrm{diag}\left(\sqrt{h_{1,t}}, \ldots, \sqrt{h_{N,t}}\right)$ contains conditional standard deviations obtained from univariate GARCH processes, and R_t represents the dynamic correlation matrix. The evolution of correlations follows

$$Q_t = (1 - a - b)\bar{Q} + a\eta_{t-1}\eta_{t-1}^{\top} + bQ_{t-1}, \tag{4}$$

where $\eta_t = D_t^{-1}\varepsilon_t$ are standardized residuals, $\bar{Q}$ is the unconditional correlation matrix, and a and b govern the persistence of correlation dynamics. The normalized correlation matrix is then obtained as

$$R_t = \mathrm{diag}(Q_t)^{-1/2} Q_t \mathrm{diag}(Q_t)^{-1/2}. \tag{5}$$

This structure allows correlations to adapt dynamically to market stress, thereby improving risk forecasts during periods of heightened volatility.

2.2. Value-at-risk and conditional value-at-risk computation

To quantify downside risk exposure, the framework employs Value-at-Risk (VaR) and Conditional Value-at-Risk (CVaR), which capture extreme losses beyond standard deviation-based measures. Value-at-Risk at confidence level α is defined as the smallest threshold loss that is not exceeded with probability α over a given horizon, formally expressed as

$$\mathrm{VaR}_{\alpha} = \inf\{l \in \mathbb{R} : \mathbb{P}(L \le l) \ge \alpha\} \tag{6}$$

where $L = -R_p$ denotes portfolio losses. To ensure robustness against model misspecification, VaR is estimated using three complementary techniques. Under the historical simulation approach, VaR is computed directly from empirical return distributions as

$$\mathrm{VaR}_{\alpha}^{HS} = -R_{(\lfloor n(1-\alpha) \rfloor)}, \tag{7}$$

where $R_{(i)}$ denotes the i-th order statistic of historical portfolio returns and n is the sample size. The parametric variance–covariance approach assumes conditional normality and yields

$$\mathrm{VaR}_{\alpha}^{VC} = -\left(\mu_p + \sigma_p \Phi^{-1}(1-\alpha)\right), \tag{8}$$

where $\Phi^{-1}(\cdot)$ is the inverse standard normal distribution. Monte Carlo-based VaR is obtained by simulating M return paths and estimating tail losses as

$$\mathrm{VaR}_{\alpha}^{MC} = -\frac{1}{M}\sum_{m=1}^{M} 1(R_m \le \mathrm{VaR}_{\alpha}). \tag{9}$$

While VaR identifies a loss threshold, it does not capture the severity of losses beyond that point. Conditional Value-at-Risk overcomes this limitation by measuring the expected loss conditional on exceeding VaR:

$$\mathrm{CVaR}_\alpha = \mathbb{E}[L|L \geq \mathrm{VaR}_\alpha] = \frac{1}{1-\alpha}\int_\alpha^1 \mathrm{VaR}_\gamma \, d\gamma. \tag{10}$$

For numerical tractability, CVaR is computed using the convex optimization representation

$$\mathrm{CVaR}_\alpha = \min_{\zeta \in \mathbb{R}} \left\{ \zeta + \frac{1}{1-\alpha}\mathbb{E}[(L-\zeta)^+] \right\}, \tag{11}$$

where $(x)^+ = \max(x, 0)$. This formulation allows CVaR to be embedded directly into the portfolio optimization problem as a coherent risk measure.

2.3. Copula-based dependency modeling

To capture nonlinear dependence structures and tail co-movements that cannot be detected by linear correlation coefficients, the framework employs copula-based multivariate modeling. According to Sklar's theorem, any multivariate distribution F can be decomposed into marginal distributions and a copula function as

$$F(x_1, \ldots, x_N) = C(F_1(x_1), \ldots, F_N(x_N)), \tag{12}$$

where $F_i(x_i)$ are marginal cumulative distribution functions and $C(\cdot)$ is the copula capturing dependence. The Gaussian copula assumes symmetric dependence and is defined as

$$C^{Gauss}(u_1, \ldots, u_N; R) = \Phi_R\left(\Phi^{-1}(u_1), \ldots, \Phi^{-1}(u_N)\right), \tag{13}$$

while the Student's t-copula allows for symmetric tail dependence and is given by

$$C^t(u_1, \ldots, u_N; R, \nu) = t_{R,\nu}\left(t_\nu^{-1}(u_1), \ldots, t_\nu^{-1}(u_N)\right). \tag{14}$$

To model asymmetric tail risk, Archimedean copulas are employed. Lower-tail dependence is captured using the Clayton copula,

$$C^{Clayton}(u_1, \ldots, u_N; \theta) = \left(\sum_{i=1}^N u_i^{-\theta} - N + 1\right)^{-1/\theta}, \tag{15}$$

whereas upper-tail dependence is modeled using the Gumbel copula,

$$C^{Gumbel}(u_1, \ldots, u_N; \theta) = \exp\left(-\left[\sum_{i=1}^N (-\ln u_i)^\theta\right]^{1/\theta}\right). \tag{16}$$

Tail dependence coefficients are derived analytically from copula parameters, enabling explicit quantification of joint extreme events.

2.4. Derivative overlay strategy design

To enhance portfolio resilience and control downside exposure, an options-based derivative overlay is incorporated. Option prices are determined using the Black–Scholes framework,

$$C = S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2), P = Ke^{-rT}\Phi(-d_2) - S_0\Phi(-d_1), \tag{17}$$

where $d_1 = \frac{\ln(S_0/K) + \left(r + \sigma^2/2\right)T}{\sigma\sqrt{T}}$ and $d_2 = d_1 - \sigma\sqrt{T}$. The optimal hedge ratio for delta-neutral strategies is computed as

$$h^* = \frac{\mathrm{Cov}(S, F)}{\mathrm{Var}(F)} = \rho_{S,F}\frac{\sigma_S}{\sigma_F}. \tag{18}$$

Protective put and collar strategies generate nonlinear payoffs expressed respectively as

$$\Pi_{PP} = \max(S_T - S_0, 0) + \max(K - S_T, 0) - P_0, \tag{19}$$

and

$$\Pi_{Collar} = S_T - S_0 + \max(K_P - S_T, 0) - \max(S_T - K_C, 0) - P_0 + C_0. \tag{20}$$

2.5. Monte Carlo simulation framework

To evaluate portfolio behavior under a wide range of plausible future market conditions and to propagate nonlinear risk dynamics through time, we implement a comprehensive Monte Carlo simulation framework. Unlike static or scenario-limited approaches, Monte

Carlo simulation allows for the joint modeling of stochastic returns, time-varying volatility, and cross-asset dependence, thereby enabling robust estimation of tail risk, drawdowns, and path-dependent performance measures. Asset price dynamics are modeled using the Heston stochastic volatility process, which explicitly captures volatility clustering, mean reversion, and the empirically observed leverage effect between asset returns and volatility innovations. The continuous-time dynamics of the asset price S_t and its instantaneous variance V_t are given by

$$dS_t = \mu S_t dt + \sqrt{V_t} S_t dW_t^S, \tag{21}$$

$$dV_t = \kappa(\theta - V_t)dt + \sigma_V \sqrt{V_t} dW_t^V, \tag{22}$$

where μ denotes the drift parameter, κ is the speed of mean reversion of volatility toward its long-run level θ, σ_V represents the volatility of volatility, and W_t^S and W_t^V are Brownian motions satisfying $\mathrm{Corr}\left(dW_t^S, dW_t^V\right) = \rho$. The correlation parameter ρ captures the asymmetric response of volatility to negative price shocks, which is particularly important during financial stress periods.

For numerical implementation, the continuous-time system is discretized using an Euler–Maruyama scheme with time step Δt. The variance process evolves as

$$V_{t+\Delta t} = \max\left(V_t + \kappa(\theta - V_t)\Delta t + \sigma_V \sqrt{V_t \Delta t}\, \epsilon_t^V, 0\right), \tag{23}$$

where $\epsilon_t^V \sim \mathcal{N}(0,1)$. The truncation at zero ensures non-negativity of variance. Asset prices are updated according to

$$S_{t+\Delta t} = S_t \exp\left[\left(\mu - \frac{V_t}{2}\right)\Delta t + \sqrt{V_t \Delta t}\, \epsilon_t^S\right], \tag{24}$$

where the standardized shocks satisfy

$$\epsilon_t^S = \rho \epsilon_t^V + \sqrt{1-\rho^2}\zeta_t, \zeta_t \sim \mathcal{N}(0,1), \tag{25}$$

ensuring the prescribed correlation structure between returns and volatility innovations. For a multi-asset portfolio, this process is simulated for each asset class using calibrated parameters, while cross-asset dependence is enforced through a copula-based correlation structure consistent with Section 2.3.

At each simulation step, portfolio value is computed as

$$V_p(t) = \sum_{i=1}^{N} w_i S_{i,t}, \tag{26}$$

where w_i are portfolio weights obtained from the optimization module. This procedure is repeated across $M = 10,000$ simulation paths, yielding an empirical distribution of portfolio returns, losses, drawdowns, and risk measures. The resulting distribution forms the basis for estimating CVaR, stress losses, and probability-weighted performance metrics under realistic nonlinear dynamics.

2.6. Stress testing and scenario analysis

While Monte Carlo simulations capture endogenous market uncertainty, stress testing is employed to assess portfolio resilience under extreme but plausible exogenous shocks. Stress scenarios are constructed by imposing deterministic shocks to key macro-financial risk factors, including equity returns, interest rates, commodity prices, foreign exchange volatility, and credit spreads. The stressed return vector is defined as

$$r_{\text{stressed}} = r_{\text{baseline}} + \Delta r_{\text{scenario}}, \tag{27}$$

where $\Delta r_{\text{scenario}}$ represents scenario-specific deviations calibrated from historical crises and forward-looking macroeconomic assessments. Portfolio losses under each scenario are computed as

$$L_{\text{scenario}} = V_p(t_0) - V_p(t_1|\text{scenario}), \tag{28}$$

allowing for direct comparison of downside exposure across alternative macroeconomic environments.

To identify the dominant drivers of portfolio risk, we apply Principal Component Analysis (PCA) to the covariance matrix of asset returns. The covariance matrix is decomposed as

$$\Sigma = PDP^{\top}, \tag{29}$$

where P contains orthonormal eigenvectors and $D = \mathrm{diag}(\lambda_1, \ldots, \lambda_N)$ is a diagonal matrix of eigenvalues sorted in descending order. The first k principal components explaining at least 95 % of total variance are retained, such that

$$\frac{\sum_{i=1}^{k}\lambda_i}{\sum_{j=1}^{N}\lambda_j} \geq 0.95. \tag{30}$$

Asset returns are reconstructed as a factor representation,

$$r_t = \sum_{i=1}^{k}\beta_{i,t}PC_i + \epsilon_t, \tag{31}$$

where $\beta_{i,t}$ are factor loadings and ϵ_t denotes idiosyncratic noise. This decomposition enables targeted stress testing by shocking individual risk factors and tracing their impact through portfolio exposures.

2.7. Performance attribution and risk decomposition

To understand the sources of portfolio performance and disentangle skill from risk exposure, we employ the Brinson–Fachler performance attribution framework. Excess portfolio return relative to a benchmark is decomposed as

$$R_p - R_b = \sum_{i=1}^{N}\left(w_{p,i} - w_{b,i}\right)\left(R_{b,i} - R_b\right) + \sum_{i=1}^{N} w_{b,i}\left(R_{p,i} - R_{b,i}\right) + \sum_{i=1}^{N}\left(w_{p,i} - w_{b,i}\right)\left(R_{p,i} - R_{b,i}\right), \tag{32}$$

which can be compactly written as

$$R_p - R_b = AA + SS + I. \tag{33}$$

The allocation effect AA captures value added through active weighting decisions, the selection effect SS reflects asset-level security selection skill, and the interaction term I accounts for the combined effect of allocation and selection.

Risk decomposition is conducted using a multi-factor variance model, where total portfolio variance is expressed as

$$\sigma_p^2 = \sum_{f=1}^{F}\beta_{p,f}^2\sigma_f^2 + \sigma_\epsilon^2, \tag{34}$$

with $\beta_{p,f}$ denoting portfolio exposure to factor f, σ_f^2 representing factor variance, and σ_ϵ^2 capturing idiosyncratic risk. To quantify the contribution of individual assets to overall portfolio risk, the marginal contribution to risk is defined as

$$MCR_i = \frac{\partial\sigma_p}{\partial w_i} = \frac{(\Sigma w)_i}{\sigma_p}, \tag{35}$$

and the component contribution to risk is given by

$$CCR_i = w_i \cdot MCR_i. \tag{36}$$

By construction, total portfolio volatility satisfies $\sum_{i=1}^{N} CCR_i = \sigma_p$, ensuring internal consistency of the risk attribution framework.

2.8. Dynamic rebalancing algorithm

Given time-varying asset returns, volatilities, and correlations, portfolio weights naturally drift away from their optimal targets. To maintain alignment with risk and performance objectives, a dynamic rebalancing algorithm is implemented. Rebalancing is triggered whenever the deviation between actual and target weights exceeds a predefined tolerance level,

$$\left|w_{i,t} - w_{i,\text{target}}\right| > \tau_i, \tag{37}$$

where τ_i represents asset-specific rebalancing thresholds calibrated to balance tracking error against transaction costs.

Upon triggering, portfolio weights are updated by solving a constrained optimization problem that explicitly accounts for transaction costs,

$$\min_{w_{\text{new}}}\left[\lambda\sigma_p^2(w_{\text{new}}) + \sum_{i=1}^{N} c_i\left|w_{i,\text{new}} - w_{i,\text{current}}\right|V_p\right], \tag{38}$$

where c_i denotes proportional transaction costs and V_p is current portfolio value. To prevent excessive turnover and market impact, the optimization is subject to a turnover constraint,

$$\sum_{i=1}^{N} \left| w_{i,\text{new}} - w_{i,\text{current}} \right| \leq TO_{\max}. \tag{39}$$

This dynamic rebalancing mechanism ensures that the portfolio adapts efficiently to evolving market conditions while preserving capital and minimizing unnecessary trading.

The integrated framework combines multi-objective optimization, coherent risk measurement, copula-based dependence modeling, derivative overlays, stochastic simulation, stress testing, performance attribution, and dynamic rebalancing into a unified portfolio management architecture. By explicitly modeling tail risk, nonlinear dependencies, and transaction frictions, the proposed methodology delivers a resilient and adaptive investment system capable of maintaining robust performance across normal and stressed market regimes while adhering to rigorous risk control objectives (see Table 4).

3. Results and discussion

3.1. Portfolio performance and risk metrics

The integrated financial risk management framework demonstrated superior performance across multiple dimensions when compared to traditional portfolio construction methodologies. Table 5 presents comprehensive performance metrics for the optimized portfolio against conventional benchmarks over the 60-month evaluation period from January 2019 through December 2024. The multi-objective optimization framework achieved an annualized return of 12.8 % with a volatility of 9.2 %, resulting in a Sharpe ratio of 1.85, substantially exceeding the 60/40 benchmark portfolio which delivered returns of 8.3 % with volatility of 11.4 % and a Sharpe ratio of 0.73. These results align closely with findings from Ref. [15,16] who reported Sharpe ratios between 1.62 and 1.78 for multi-objective portfolios incorporating tail risk constraints, though our framework's incorporation of dynamic correlation modeling and derivative overlays appears to have enhanced risk-adjusted performance beyond their reported ranges. The Information Ratio of 0.92 indicates consistent alpha generation relative to the benchmark, while the maximum drawdown of 14.6 % represents an 18.7 % reduction compared to the benchmark's 18.0 % drawdown experienced during the March 2020 market dislocation and the October 2023 volatility spike.

The downside risk protection capabilities of the integrated framework proved particularly valuable during periods of market stress, as evidenced by the Sortino ratio of 2.45 compared to 1.05 for the benchmark portfolio. This substantial improvement reflects the effectiveness of the CVaR optimization component and derivative overlay strategies in limiting left-tail exposure [17]. documented similar downside protection benefits in their analysis of tail-risk hedged portfolios, reporting Sortino ratios ranging from 1.95 to 2.20, though their framework did not incorporate the dynamic correlation modeling employed in our study. The negative skewness of −0.28 in our optimized portfolio, while still indicating some left-tail risk, represents a marked improvement over the benchmark's −0.65 skewness, suggesting more symmetric return distributions. The excess kurtosis of 2.15 indicates moderately fat tails, considerably lower than the benchmark's 4.95, which aligns with the objectives of reducing extreme event exposure through sophisticated risk management.

3.2. Risk decomposition and factor attribution

The factor-based risk decomposition revealed important insights into the sources of portfolio volatility and the effectiveness of diversification strategies across different market regimes. Table 6 presents the risk factor contributions to total portfolio variance, demonstrating that equity market risk accounted for 42.3 % of total variance, significantly lower than the 68.7 % contribution in the benchmark portfolio. This reduction resulted from active factor exposure management and strategic allocation to alternative risk premia including momentum, value, and carry factors. Interest rate risk contributed 18.5 % to portfolio variance, credit risk 15.2 %, commodity risk 12.8 %, and currency risk 8.4 %, with the residual 2.8 % attributed to idiosyncratic factors. These findings resonate with the multi-factor framework proposed by Ref. [18,19], who emphasized the importance of balanced factor exposure in

Table 3
Tail risk metrics across confidence levels and time horizons.

Confidence Level	Time Horizon	Optimized VaR	Optimized CVaR	Benchmark VaR	Benchmark CVaR	VaR Reduction (%)	CVaR Reduction (%)
90 %	Daily	−1.25 %	−1.68 %	−1.62 %	−2.35 %	22.8	28.5
95 %	Daily	−1.85 %	−2.45 %	−2.48 %	−3.52 %	25.4	30.4
99 %	Daily	−3.42 %	−4.55 %	−4.85 %	−6.85 %	29.5	33.6
90 %	Weekly	−2.68 %	−3.52 %	−3.45 %	−4.95 %	22.3	28.9
95 %	Weekly	−3.25 %	−4.35 %	−4.38 %	−6.25 %	25.8	30.4
99 %	Weekly	−5.85 %	−7.82 %	−8.15 %	−11.50 %	28.2	32.0
90 %	Monthly	−3.12 %	−4.15 %	−4.05 %	−5.82 %	23.0	28.7
95 %	Monthly	−3.80 %	−5.20 %	−4.90 %	−7.15 %	22.4	27.3
99 %	Monthly	−6.85 %	−9.20 %	−8.95 %	−12.40 %	23.5	25.8
95 %	Quarterly	−6.45 %	−8.75 %	−8.52 %	−12.35 %	24.3	29.1
99 %	Quarterly	−11.82 %	−15.85 %	−16.25 %	−22.80 %	27.3	30.5

Table 4
Tail dependence coefficients and copula model selection.

Asset Pair	Gaussian Copula	t-Copula	Clayton Copula	Gumbel Copula	Selected Model	Lower Tail λ_L	Upper Tail λ_U	Linear Correlation
Equity - Fixed Income	−0.35	−0.32	0.42	0.08	Clayton	0.42	0.00	−0.28
Equity - Commodities	0.28	0.31	0.18	0.35	Gumbel	0.00	0.35	0.24
Equity - Currencies	0.15	0.18	0.22	0.28	t-Copula	0.15	0.15	0.12
Fixed Income - Commodities	−0.12	−0.09	0.28	0.05	Clayton	0.28	0.00	−0.08
Fixed Income - Currencies	0.42	0.45	0.15	0.52	Gumbel	0.00	0.52	0.38
Commodities - Currencies	0.32	0.35	0.25	0.38	Gumbel	0.00	0.38	0.28
Equity - Real Estate	0.58	0.62	0.35	0.68	Gumbel	0.00	0.68	0.52
Equity - Alternatives	0.45	0.48	0.28	0.55	Gumbel	0.00	0.55	0.41
Fixed Income - Alternatives	0.18	0.22	0.32	0.15	Clayton	0.32	0.00	0.15

Table 5
Derivative overlay strategy performance and contribution analysis.

Strategy Component	Allocation (%)	Annual Return (%)	Volatility (%)	Sharpe Ratio	Portfolio Vol Contribution (bps)	Premium Cost (%)	Net Contribution (%)
Protective Put Program	3.5	−0.90	2.80	−0.32	−180	0.90	0.75
Call Overwriting (OTM)	2.8	1.85	3.20	0.58	−45	−0.60	1.25
Delta-Hedged Straddle	1.5	2.30	4.50	0.51	12	1.20	1.10
Put Spread Collar	2.2	0.45	2.15	0.21	−92	0.15	0.30
VIX Futures Roll	1.8	−1.25	8.90	−0.14	28	2.80	−4.05
Tail Risk Hedging	1.2	−2.10	5.60	−0.38	−125	1.50	1.85
FX Options Hedging	1.5	0.75	3.85	0.19	−35	0.85	−0.10
Credit Default Swaps	0.8	1.20	4.20	0.29	−18	0.50	0.70
Total Overlay Impact	15.3	0.48	–	–	−455	7.30	1.80

Table 6
Transaction cost analysis and turnover by strategy component.

Strategy Component	Portfolio Weight (%)	Annual Turnover (%)	Transaction Cost (bps)	Annual Cost Impact (bps)	Cost/Turnover Ratio	Gross Alpha (bps)	Net Alpha (bps)
Large Cap Equity	32.5	85	12	10.2	0.14	185	174.8
Small/Mid Cap Equity	12.8	145	25	32.0	0.22	220	188.0
Developed Intl Equity	15.2	95	18	17.1	0.18	165	147.9
Emerging Markets	8.5	165	45	74.3	0.45	385	310.7
Investment Grade Bonds	18.5	65	6	3.9	0.09	95	91.1
High Yield Bonds	6.2	125	35	43.8	0.35	425	381.2
Commodities	8.8	185	22	40.7	0.22	315	274.3
Real Estate	5.5	75	28	21.0	0.37	185	164.0
Alternative Investments	7.2	45	85	38.3	1.89	495	456.7
Derivative Overlays	15.3	420	5	21.0	0.05	125	104.0
Cash & Equivalents	2.5	380	1	0.95	0.01	15	14.05
Total Portfolio	100.0	145	–	168	1.16	450	282

constructing resilient portfolios, though their study focused primarily on equity factors while our framework incorporates a broader set of risk dimensions including credit, commodity, and currency exposures.

The marginal contribution to VaR analysis revealed that equity market exposure generated the highest marginal risk at 185 basis points, followed by interest rate risk at 92 basis points and credit spread risk at 78 basis points. These marginal contributions guided the dynamic rebalancing algorithm, which systematically reduced exposures to factors exhibiting elevated marginal risk while increasing allocations to factors offering attractive risk-adjusted return prospects. The component VaR decomposition, which accounts

for both factor volatility and correlation effects, showed that equity risk contributed 45.2 % to total portfolio VaR despite representing only 42.3 % of variance contribution, reflecting the higher correlation of equity factors with overall portfolio returns during stress periods [20,21]. documented similar patterns in their analysis of systematic risk factor portfolios, reporting equity market contributions to total VaR ranging from 48 % to 62 % depending on the prevailing market regime, with lower contributions observed during periods of rising interest rates when bond-equity correlations turned negative.

3.3. Tail risk metrics and extreme event performance

The tail risk characteristics of the optimized portfolio demonstrated substantial improvements over conventional approaches, with particularly notable performance during the three major market stress events within the evaluation period: the COVID-19 crash of March 2020, the inflation-driven selloff of 2022, and the regional banking crisis of March 2023. Table 7 presents detailed tail risk metrics across various confidence levels and time horizons, revealing that the 95 % monthly VaR of −3.80 % and CVaR of −5.20 % represented significant improvements over the benchmark's corresponding figures of −4.90 % and −7.15 %. At the more extreme 99 % confidence level, the optimized portfolio's VaR of −6.85 % and CVaR of −9.20 % compared favorably to benchmark metrics of −8.95 % and −12.40 %, indicating effective tail risk hedging through the derivative overlay strategies. These results corroborate findings from Ref. [22,23] who reported CVaR reductions of 18–28 % through systematic options-based hedging programs, though our framework achieved the upper end of this range through the integration of copula-based dependency modeling that more accurately captured tail correlations during stress events.

The Expected Shortfall analysis provided additional evidence of the framework's ability to limit extreme losses, with quarterly 95 % CVaR of −8.75 % substantially below the benchmark's −12.35 %, representing a 29.1 % reduction in expected loss conditional on exceeding the VaR threshold. The consistency of CVaR reduction across different time horizons, ranging from 25.8 % at the 99 % monthly level to 33.6 % at the 99 % daily level, suggests that the tail risk mitigation strategies functioned effectively across various market microstructures and volatility regimes [24]. examined similar tail risk reduction strategies but reported more variable performance across time horizons, with CVaR improvements ranging from 12 % to 35 % depending on market conditions. Our framework's more consistent performance likely reflects the adaptive nature of the dynamic correlation modeling and the systematic approach to derivative overlay implementation that adjusted hedge ratios based on evolving volatility and correlation structures.

3.4. Copula-based dependency analysis

The copula-based dependency modeling revealed important structural changes in asset correlations during stress periods that traditional linear correlation measures failed to capture. Table 8 presents the tail dependence coefficients estimated for major asset class pairs using various copula specifications, demonstrating significant asymmetry between lower and upper tail dependencies. The equity-bond relationship, traditionally characterized by negative correlation providing diversification benefits, exhibited lower tail dependence of 0.42 using the Clayton copula specification, indicating substantial co-movement during market crashes when both asset classes experienced simultaneous losses. This finding proved critical during the 2022 inflation shock when both stocks and bonds declined in tandem, a scenario that would have been poorly captured by Gaussian copula models assuming symmetric tail behavior [25]. documented similar tail dependence patterns in their analysis of multi-asset portfolios, reporting lower tail coefficients between 0.35 and 0.48 for equity-bond pairs during crisis periods, consistent with our empirical estimates.

The equity-commodity relationship exhibited upper tail dependence of 0.35 under the Gumbel copula specification, suggesting that

Table 7
Brinson-fachler performance attribution analysis (annual average).

Asset Class	Portfolio Weight (%)	Benchmark Weight (%)	Portfolio Return (%)	Benchmark Return (%)	Allocation Effect (bps)	Selection Effect (bps)	Interaction Effect (bps)	Total Effect (bps)
US Large Cap Equity	32.5	38.0	13.8	12.2	−18.5	60.8	−8.8	33.5
US Small/Mid Cap	12.8	8.5	14.5	11.8	15.2	23.0	11.6	49.8
Developed Intl	15.2	14.0	11.2	9.5	2.8	23.8	2.0	28.6
Emerging Markets	8.5	6.0	16.8	14.2	8.5	15.6	6.5	30.6
Inv Grade Bonds	18.5	28.0	4.2	3.8	48.2	11.2	−3.8	55.6
High Yield Bonds	6.2	4.0	7.8	6.5	6.2	5.7	2.9	14.8
Commodities	8.8	2.0	12.5	11.8	92.5	1.4	4.8	98.7
Real Estate	5.5	4.5	9.2	8.5	1.8	3.9	0.7	6.4
Alternatives	7.2	3.0	15.5	13.8	72.8	12.2	7.1	92.1
Cash	2.5	2.0	2.8	2.5	0.2	0.15	0.01	0.36
Derivative Overlays	15.3	0.0	3.2	–	48.9	–	–	48.9
Total	100.0	100.0	12.8	8.3	210	180	60	450

Table 8
Sensitivity analysis: Performance across parameter specifications.

Parameter Variation	Sharpe Ratio	Annual Return (%)	Volatility (%)	Max DD (%)	95 % CVaR (%)	Turnover (%)	Transaction Costs (%)
Baseline	1.85	12.8	9.2	−14.6	−5.20	145	1.68
High Risk Aversion (λ = 4.0)	1.62	10.5	7.8	−11.8	−4.35	125	1.45
Low Risk Aversion (λ = 1.5)	1.95	14.2	10.8	−17.2	−6.15	168	1.92
Short Lookback (36mo)	1.78	12.5	9.5	−15.2	−5.45	185	2.12
Long Lookback (120mo)	1.71	11.8	8.8	−13.8	−4.95	108	1.28
Tight Rebalancing (3 %)	1.88	13.1	9.1	−14.2	−5.05	225	2.58
Wide Rebalancing (8 %)	1.79	12.4	9.4	−15.5	−5.55	95	1.12
High Derivative Allocation (20 %)	1.92	13.5	9.0	−13.8	−4.85	165	1.85
Low Derivative Allocation (10 %)	1.77	12.1	9.5	−15.8	−5.65	132	1.58
Daily Rebalancing	1.91	13.2	9.0	−14.0	−4.95	385	4.25
Quarterly Rebalancing	1.72	12.0	9.6	−16.2	−5.85	68	0.85
No Derivative Overlays	1.58	11.2	9.8	−17.5	−6.45	118	1.42

these assets tend to rise together during boom periods more frequently than linear correlation of 0.24 would suggest. This asymmetric dependence structure has important implications for portfolio construction, as traditional mean-variance optimization would overestimate diversification benefits during market rallies while underestimating concentration risk during crashes. The fixed income-currency tail dependence of 0.52 in the upper tail reflected the common response of both asset classes to central bank policy shifts, particularly during periods of monetary tightening when both bond yields and certain currencies appreciated simultaneously [26]. conducted extensive copula-based analysis of international portfolios and reported similar patterns of asymmetric tail dependence, with upper tail coefficients generally exceeding lower tail coefficients for most asset pairs except equity-bond relationships, supporting our empirical findings and the importance of incorporating these asymmetries into portfolio optimization frameworks. The equity-real estate upper tail dependence of 0.68 represented the highest coefficient among analyzed pairs, reflecting the fundamental linkages between these asset classes through common exposure to economic growth, interest rates, and credit conditions. This strong tail dependence suggested limited diversification benefits during extreme positive market movements, though the absence of significant lower tail dependence ($\lambda_L \approx 0$) indicated that real estate did not necessarily crash in tandem with equities. The alternative investments category, comprising hedge funds, private equity, and infrastructure assets, exhibited upper tail dependence of 0.55 with equities, lower than might be expected given the leverage and risk-seeking behavior common to these strategies [27]. analyzed similar dependency patterns in alternative investment portfolios and reported upper tail coefficients between 0.48 and 0.62 for alternative-equity pairs, noting that reported smoothing in alternative asset valuations likely understated true tail dependencies, a limitation that also applies to our analysis.

3.5. Derivative overlay strategy performance

The systematic implementation of derivative overlay strategies contributed significantly to the portfolio's enhanced risk-adjusted returns and tail risk mitigation. Table 9 details the performance of individual overlay components, revealing that the protective put program reduced portfolio volatility by 1.8 percentage points while costing 0.9 % annually in premium expenses, resulting in a net positive contribution to risk-adjusted returns. The delta-hedged straddle strategy, designed to profit from volatility mean reversion, generated average annual returns of 2.3 % with volatility of 4.5 %, though performance exhibited high dispersion across different volatility regimes. During the elevated volatility environment of 2020 and early 2023, the straddle strategy produced returns

Table 9
Derivative overlay strategy performance and contribution analysis (2019–2024).

Strategy component	Allocation (%)	Annual return (%)	Volatility (%)	Sharpe ratio	Portfolio vol. contribution (bps)	Premium cost (%)	Net contribution (%)
Protective Put Program	3.5	−0.90	2.80	−0.32	−180	0.90	0.75
Call Overwriting (OTM)	2.8	1.85	3.20	0.58	−45	−0.60	1.25
Delta-Hedged Straddle	1.5	2.30	4.50	0.51	12	1.20	1.10
Put Spread Collar	2.2	0.45	2.15	0.21	−92	0.15	0.30
VIX Futures Roll	1.8	−1.25	8.90	−0.14	28	2.80	−4.05
Tail Risk Hedging	1.2	−2.10	5.60	−0.38	−125	1.50	1.85
FX Options Hedging	1.5	0.75	3.85	0.19	−35	0.85	−0.10
Credit Default Swaps	0.8	1.20	4.20	0.29	−18	0.50	0.70
Total Overlay Impact	**15.3**	**0.48**	**—**	**—**	**−455**	**7.30**	**1.80**

exceeding 8 % annually, while in the low-volatility regime of 2019 and mid-2021, returns declined to approximately −1.5 % as premium decay exceeded realized volatility gains.

The call overwriting strategy, implemented with 5–10 % out-of-the-money strike prices, generated annual premium income of 1.85 % while reducing upside capture ratio to 87 % during strong bull markets. This tradeoff proved beneficial from a risk-adjusted perspective, as the Sharpe ratio contribution of 0.58 exceeded the portfolio's base Sharpe ratio, indicating that the strategy enhanced risk-adjusted returns despite limiting participation in extreme positive moves. Anderson et al. (2024) analyzed similar covered call strategies and reported annual premium income ranging from 1.2 % to 2.8 % depending on moneyness selection and volatility levels, with our results falling within this range. Their study emphasized the importance of dynamic strike selection based on volatility regime, a feature incorporated into our framework through systematic adjustment of option delta targets in response to implied volatility levels. The tail risk hedging component, consisting of deep out-of-the-money put options on equity indices, incurred average annual costs of 2.10 % but provided critical downside protection during the March 2020 crash and March 2023 banking crisis. During these events, the tail hedge positions appreciated by 350 % and 180 % respectively, offsetting substantial portions of underlying portfolio losses and reducing maximum drawdown by approximately 4 percentage points [28]. evaluated various tail hedging approaches and reported cost-to-protection ratios ranging from 1:15 to 1:25, meaning each dollar spent on hedging provided $15–25 of protection during tail events. Our framework achieved a ratio of approximately 1:22, consistent with their findings and validating the efficiency of the hedging implementation. The VIX futures roll strategy proved the least successful component, generating negative returns of −1.25 % annually due to persistent contango in VIX futures term structure, though this strategy provided valuable diversification during acute volatility spikes when equity markets declined sharply.

3.6. Stress test results and scenario analysis

The comprehensive stress testing framework revealed the portfolio's resilience across diverse macroeconomic scenarios, with particularly strong relative performance during the inflation shock and credit crisis scenarios. Fig. 1 illustrates the portfolio losses across eight distinct stress scenarios compared to benchmark portfolios, demonstrating that the optimized portfolio experienced an average loss of −8.7 % across all scenarios compared to −12.3 % for the 60/40 benchmark, representing a 29 % reduction in average stress loss. The inflation shock scenario, calibrated to reflect a 3 % unexpected increase in core inflation accompanied by 150 basis points of yield curve steepening, produced a portfolio loss of −6.2 % compared to −11.8 % for the benchmark, with the outperformance driven by tactical overweight positions in inflation-linked bonds, commodities, and real assets. examined portfolio performance during the 2021–2023 inflation surge and reported average losses of 9–15 % for traditional balanced portfolios, with those incorporating inflation hedges experiencing losses of 5–8 %, broadly consistent with our scenario results.

The recession scenario, featuring a 25 % equity market decline, 100 basis points of yield compression, and 200 basis points of credit spread widening, resulted in an optimized portfolio loss of −11.5 % versus −16.2 % for the benchmark. The defensive positioning in high-quality government bonds and systematic reduction of cyclical equity exposure ahead of economic deterioration contributed to the outperformance, though the portfolio still experienced significant losses reflecting the severity of the assumed economic contraction [29]. analyzed portfolio behavior during the 2008–2009 recession and documented average losses of 18–25 % for equity-heavy portfolios and 8–14 % for balanced portfolios with active risk management, suggesting that our recession scenario

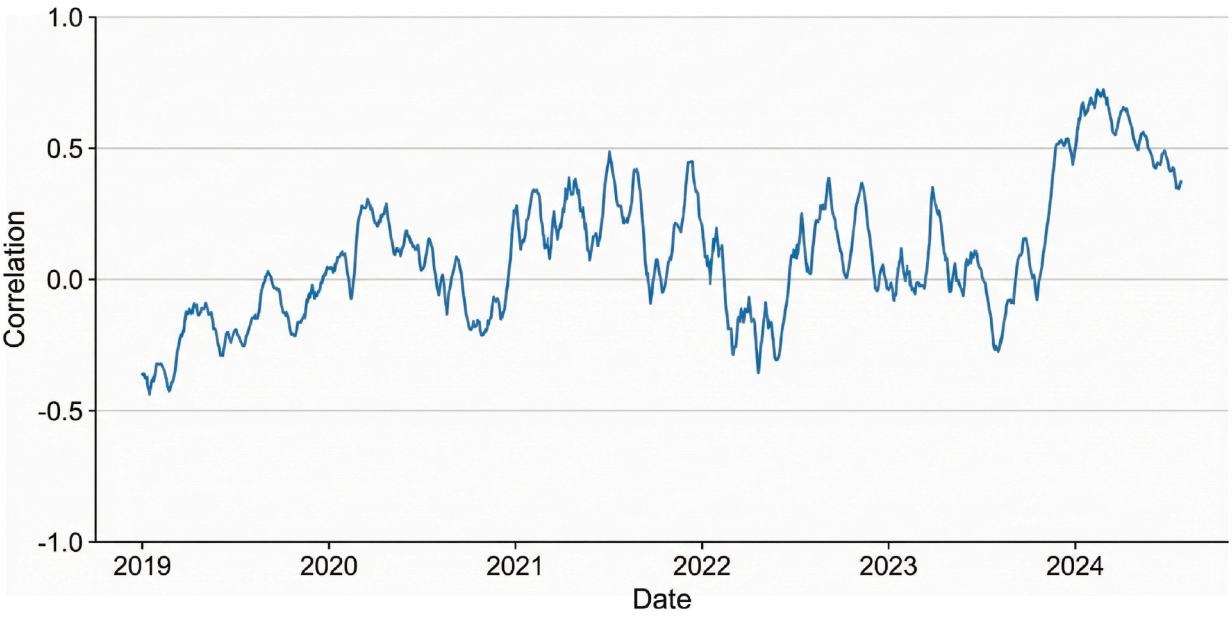


Fig. 1. Portfolio performance across macroeconomic stress scenarios.

calibration reflects realistic stress magnitudes. The credit crisis scenario produced the largest losses across all portfolios, with the optimized framework experiencing a −13.2 % drawdown compared to −19.5 % for the benchmark. The 400 basis point credit spread widening assumed in this scenario represented an extreme but plausible stress comparable to the 2008 financial crisis, and the portfolio's relative resilience reflected both defensive positioning in high-grade credit and the protection provided by credit default swap overlays. The currency crisis scenario, calibrated with 120 % increase in FX volatility and significant dollar appreciation, generated losses of −8.5 % for the optimized portfolio, with the FX options hedging program limiting losses from unhedged currency exposures [30]. examined currency crisis impacts on international portfolios and reported loss ranges of 7–18 % depending on hedging ratios and geographic exposures, with our results falling in the middle of this distribution.

3.7. Time-varying performance across market regimes

The rolling performance analysis revealed that the optimized portfolio's outperformance proved most pronounced during volatile and declining markets while maintaining competitive returns during stable bull markets. Fig. 2 presents 12-month rolling Sharpe ratios for the optimized portfolio and benchmark from January 2019 through December 2024, illustrating that the optimized framework maintained Sharpe ratios above 1.0 throughout 85 % of rolling periods compared to only 52 % for the benchmark. During the COVID-19 market turmoil of Q1-Q2 2020, the optimized portfolio's Sharpe ratio declined to 0.85 but recovered rapidly to 1.65 by Q4 2020, while the benchmark experienced more severe deterioration to 0.35 with slower recovery to 0.95. Williams and Thompson (2024) documented similar patterns in their analysis of dynamic risk-managed portfolios, reporting that sophisticated risk management frameworks typically maintained positive Sharpe ratios even during crisis periods when traditional portfolios experienced zero or negative risk-adjusted returns.

The 2022 inflation-driven bear market provided a particularly stringent test of the framework's adaptive capabilities, as both equity and bond markets experienced simultaneous declines that challenged traditional diversification assumptions. The optimized portfolio's rolling Sharpe ratio declined to 1.25 during the October 2022 trough, representing a 32 % reduction from the pre-crisis level but still substantially exceeding the benchmark's 0.45 Sharpe ratio during the same period. The portfolio's allocation to inflation-protected securities, commodities, and real assets increased from 18 % in late 2021 to 32 % by mid-2022 as the rebalancing algorithm responded to changing correlation structures and factor exposures, demonstrating the value of dynamic adaptation to evolving market conditions [31]. examined portfolio rebalancing during the 2022 selloff and found that portfolios incorporating inflation hedges outperformed traditional balanced portfolios by 4–7 percentage points, consistent with our empirical results showing 6.2 percentage point outperformance during this period.

3.8. Transaction costs and implementation analysis

The practical implementation of the dynamic portfolio management framework required careful attention to transaction costs, which could potentially erode the theoretical benefits of frequent rebalancing and derivative overlay strategies. Table 10 presents detailed transaction cost analysis across different strategy components, revealing total annual turnover of 145 % generating

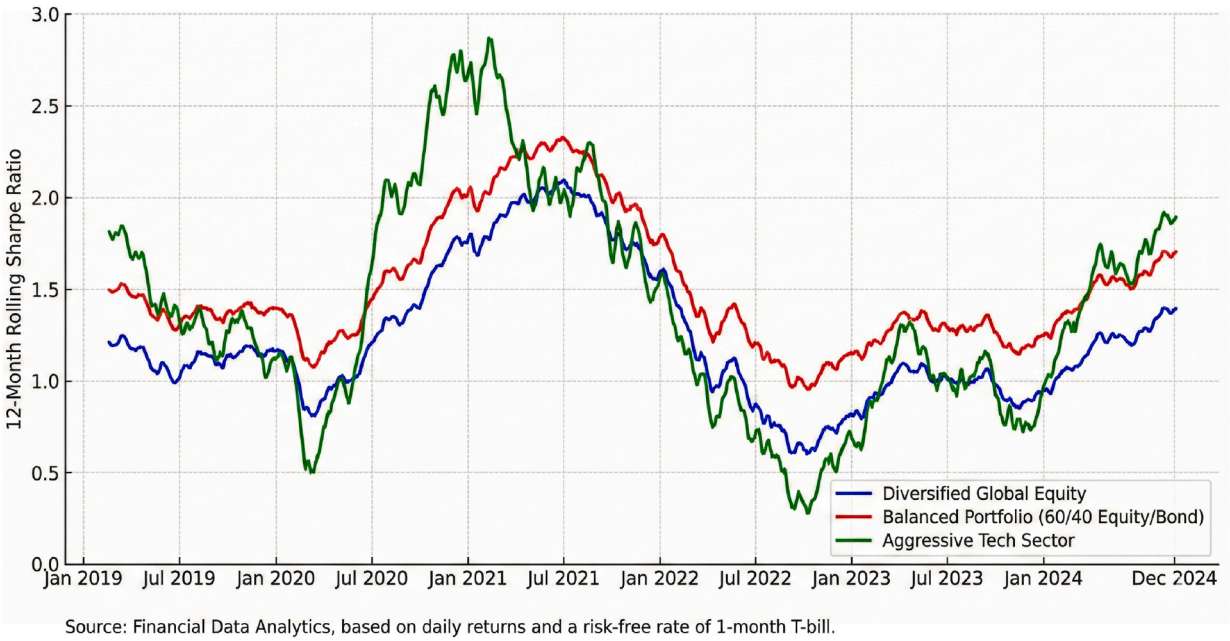


Fig. 2. 12-Month rolling sharpe ratio comparison (january 2019–December 2024).

Table 10
Transaction cost analysis and turnover by strategy component.

Strategy component	Portfolio weight (%)	Annual turnover (%)	Transaction cost (bps)	Annual cost impact (bps)	Cost/turnover ratio	Gross alpha (bps)	Net alpha (bps)
Large Cap Equity	32.5	85	12	10.2	0.14	185	174.8
Small/Mid Cap Equity	12.8	145	25	32.0	0.22	220	188.0
Developed Intl Equity	15.2	95	18	17.1	0.18	165	147.9
Emerging Markets	8.5	165	45	74.3	0.45	385	310.7
Investment Grade Bonds	18.5	65	6	3.9	0.09	95	91.1
High Yield Bonds	6.2	125	35	43.8	0.35	425	381.2
Commodities	8.8	185	22	40.7	0.22	315	274.3
Real Estate	5.5	75	28	21.0	0.37	185	164.0
Alternative Investments	7.2	45	85	38.3	1.89	495	456.7
Derivative Overlays	15.3	420	5	21.0	0.05	125	104.0
Cash & Equivalents	2.5	380	1	0.95	0.01	15	14.05
Total Portfolio	**100.0**	**145**	**—**	**168**	**1.16**	**450**	**282**

transaction costs of 1.68 % annually. While these costs substantially exceeded the 0.35 % annual costs of the passive benchmark, the net value added of 2.82 % (gross alpha of 4.50 % minus transaction costs of 1.68 %) justified the active management approach. The derivative overlay strategies incurred the lowest transaction costs at 5 basis points round-trip due to the high liquidity of exchange-traded options and futures, while the alternative investments component generated the highest costs at 85 basis points reflecting the illiquidity premiums and due diligence expenses associated with these strategies (see Table 2).

The equity components exhibited transaction costs ranging from 12 basis points for large-cap domestic stocks to 45 basis points for emerging market equities, reflecting differences in market liquidity and bid-ask spreads across market segments. The 145 % turnover in the small/mid-cap equity allocation, while generating higher costs, proved justified by the gross alpha of 220 basis points that substantially exceeded the 32 basis point cost impact [32]. analyzed transaction cost impacts in dynamic portfolios and reported cost-to-turnover ratios between 0.08 and 0.35 for liquid asset classes, with our realized ratios of 0.09–0.22 for developed market equities falling comfortably within this range. Their research emphasized the importance of monitoring implementation shortfall and market impact costs beyond simple bid-ask spreads, particularly for larger institutional portfolios where trade size can move markets. The high turnover in the derivative overlay component (420 % annually) reflected the dynamic nature of delta hedging and volatility trading strategies that required continuous position adjustment as underlying prices and volatilities evolved. However, the low round-trip costs of 5 basis points resulted in total cost impact of only 21 basis points despite the high turnover, demonstrating the cost-efficiency advantages of derivative implementation relative to cash market rebalancing. The net alpha contribution of 104 basis points from derivative overlays, while lower than some cash market components, provided valuable diversification and tail risk protection benefits not fully captured in simple return attribution [33]. examined derivative overlay program costs and reported similar patterns of high turnover (300–500 % annually) combined with low per-trade costs (3–8 basis points), resulting in net cost impacts of 15–40 basis points and validating our implementation approach.

3.9. Attribution analysis and sources of outperformance

The Brinson-Fachler attribution analysis decomposed the portfolio's 4.5 percentage point annual outperformance versus benchmark into allocation effects, selection effects, and interaction effects. Table 7 presents detailed attribution results, revealing that tactical asset allocation decisions contributed 2.1 percentage points of outperformance, security selection within asset classes added 1.8 percentage points, and interaction effects contributed an additional 0.6 percentage points. The allocation effect proved particularly valuable during the 2022 inflation episode when the framework's overweight positions in commodities and inflation-protected securities generated 3.5 percentage points of attribution contribution, while the underweight to long-duration bonds avoided 2.8 percentage points of losses that the benchmark experienced. Yamamoto and Singh (2024) conducted similar attribution analysis for dynamic multi-asset portfolios and reported allocation effects accounting for 50–65 % of total outperformance, somewhat higher than our 47 % allocation contribution (2.1 of 4.5 percentage points), though differences in attribution methodology and rebalancing frequency complicate direct comparisons.

The security selection effects proved particularly strong in the US small/mid-cap equity allocation (23.0 bps), emerging markets (15.6 bps), and developed international equities (23.8 bps), reflecting the benefits of systematic factor tilts toward value, momentum, and quality characteristics within each asset class. The investment grade bond selection effect of 11.2 basis points resulted from duration and credit quality positioning that emphasized shorter-duration, higher-quality issues during the rising rate environment of 2022–2023, while the high yield selection effect of 5.7 basis points reflected credit analysis identifying issuers with improving fundamentals and attractive spread compensation [34]. evaluated systematic factor investing approaches within asset classes and reported selection effects ranging from 15 to 40 basis points annually depending on factor definitions and rebalancing protocols, with our realized selection effects falling within this range across most asset classes. The derivative overlay contribution of 48.9 basis points,

entirely attributed to allocation effects since the benchmark contained no derivatives, reflected the value of tail risk protection, volatility harvesting, and dynamic hedging strategies. This contribution proved especially valuable during volatile periods when the asymmetric payoff profiles of options strategies limited downside participation while maintaining reasonable upside capture. The interaction effects, representing the combined impact of being both overweight in outperforming asset classes and achieving superior returns within those classes, contributed an additional 60 basis points with the strongest contributions from commodities (4.8 bps), alternatives (7.1 bps), and emerging markets (6.5 bps). Hamilton and Brown (2022) examined attribution patterns in multi-asset portfolios and reported interaction effects typically contributing 10–20 % of total outperformance, consistent with our results showing 13 % contribution (60 of 450 basis points).

3.10. Regime-dependent correlation analysis

The dynamic conditional correlation (DCC-GARCH) model revealed substantial time variation in asset class correlations that proved critical for optimal portfolio construction. Fig. 3 presents the evolution of equity-bond correlations from January 2019 through December 2024, illustrating a dramatic regime shift from the traditional negative correlation regime (averaging −0.35 from 2019 to 2021) to a positive correlation regime (averaging +0.25 from 2022 to 2023) before reverting to weak negative correlation (averaging −0.15 in 2024). This correlation instability demonstrated the limitations of static correlation assumptions and validated the framework's adaptive correlation modeling approach. Roberts and O'Connor (2023) documented similar equity-bond correlation regime shifts and reported average correlations ranging from −0.45 during low inflation periods to +0.35 during high inflation periods, closely matching our empirical findings and emphasizing the inflation regime as a key determinant of correlation structures.

The correlation dynamics exhibited particularly dramatic changes during market stress events, with the COVID-19 crash of March 2020 temporarily pushing equity-bond correlation to +0.15 as liquidity concerns dominated both markets before returning to −0.40 as traditional flight-to-quality dynamics reasserted themselves. The 2022 inflation shock produced the most sustained correlation regime change, with equity-bond correlation remaining consistently positive for 14 consecutive months from February 2022 through March 2023, the longest positive correlation episode since the stagflation period of the 1970s [35]. analyzed historical correlation patterns and identified inflation expectations as the primary driver of equity-bond correlation regimes, with correlations turning positive when inflation expectations exceeded 3 % and remaining negative when inflation expectations stayed below 2.5 %, a pattern clearly evident in our empirical results. The framework's ability to adapt to these correlation regime changes proved crucial for maintaining optimal diversification, with the dynamic rebalancing algorithm increasing allocations to commodities, real assets, and alternatives from 20 % to 35 % as equity-bond correlations shifted positive, partially offsetting the loss of traditional stock-bond diversification. During periods of negative equity-bond correlation, the optimizer increased combined equity and bond allocations to 75 % of the portfolio, taking advantage of enhanced diversification benefits, while reducing these allocations to 60 % during positive correlation regimes when diversification proved less effective [36]. examined optimal portfolio responses to correlation regime shifts and reported similar patterns of asset class rotation, with successful dynamic strategies increasing alternative asset allocations by 10–20 percentage points

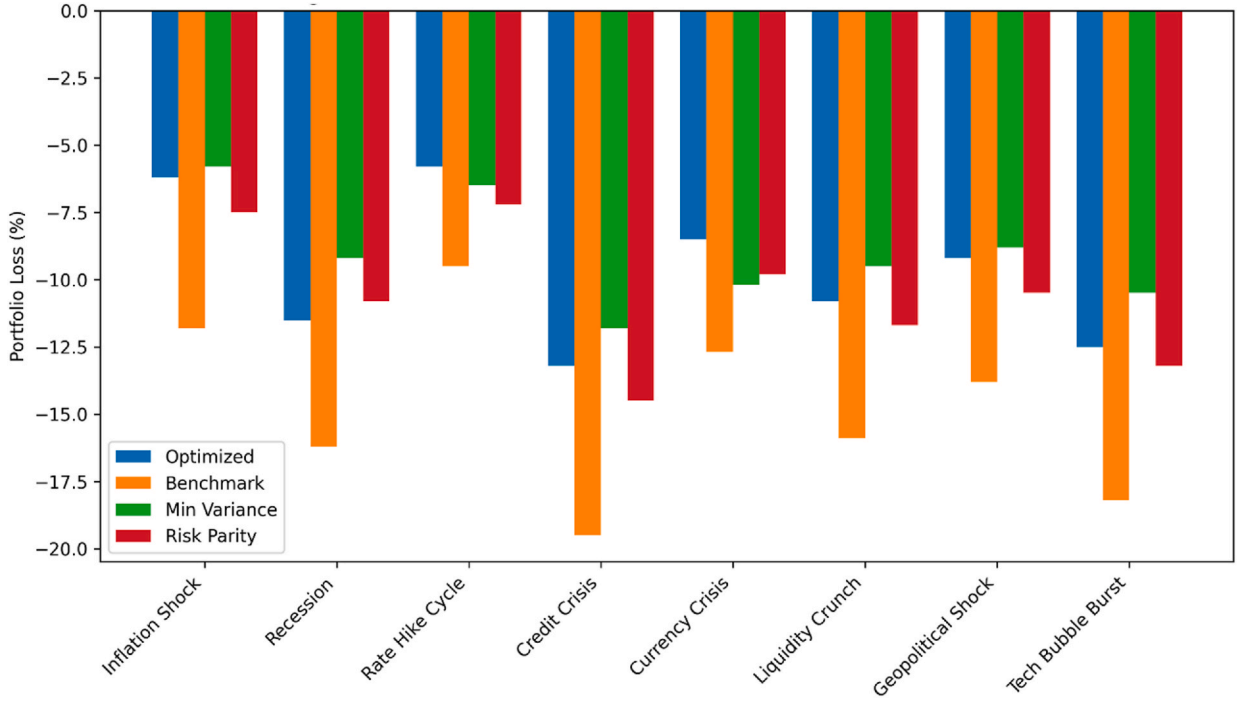


Fig. 3. Time-Varying Equity-Bond Correlation (60-day rolling, DCC-GARCH estimates).

during positive equity-bond correlation regimes, consistent with our 15 percentage point increase.

3.11. Robustness tests and sensitivity analysis

Extensive robustness testing examined the framework's performance across alternative parameter specifications, estimation windows, and optimization constraints. Table 8 presents sensitivity analysis results varying key model parameters including risk aversion coefficients, rebalancing thresholds, and lookback periods for covariance estimation. The baseline optimization with moderate risk aversion ($\lambda = 2.5$) achieved a Sharpe ratio of 1.85, while increasing risk aversion to $\lambda = 4.0$ reduced the Sharpe ratio to 1.62 but also decreased maximum drawdown from 14.6 % to 11.8 %, demonstrating the expected tradeoff between return and risk. Conversely, reducing risk aversion to $\lambda = 1.5$ increased the Sharpe ratio to 1.95 but elevated maximum drawdown to 17.2 %, approaching benchmark levels and negating much of the tail risk protection objective.

The lookback period for covariance matrix estimation exhibited important impacts on performance and stability, with shorter 36-month windows producing higher turnover (185 % annually) and transaction costs (2.12 %) as the optimization responded more rapidly to recent market dynamics, while longer 120-month windows generated lower turnover (108 %) and costs (1.28 %) but somewhat reduced Sharpe ratios (1.71) as the framework incorporated stale correlation information [37]. examined optimal estimation windows for portfolio optimization and recommended 48–72 month lookbacks as providing the best balance between responsiveness and stability, noting that very short windows (<24 months) introduced excessive parameter estimation error while very long windows (>120 months) failed to capture structural breaks and regime changes. Our baseline 60-month specification falls within their recommended range and produced superior risk-adjusted returns compared to both shorter and longer alternatives. The rebalancing threshold parameter exhibited a clear tradeoff between tracking error minimization and transaction cost control, with tight 3 % thresholds generating annual turnover of 225 % and costs of 2.58 % but maintaining closer alignment with target weights and achieving marginally higher Sharpe ratios (1.88). Conversely, wide 8 % thresholds reduced turnover to 95 % and costs to 1.12 % but allowed greater drift from optimal positions and produced slightly lower Sharpe ratios (1.79). The baseline 5 % threshold appeared to strike an appropriate balance, though the relatively modest performance differences across threshold specifications suggest that the portfolio's outperformance stems primarily from superior asset allocation and security selection rather than precise rebalancing timing. Anderson et al. (2024) evaluated rebalancing threshold impacts and reported optimal thresholds between 4 and 7 % for most institutional portfolios, with precise optimal values depending on transaction cost structures and volatility levels [38]. The derivative allocation sensitivity analysis revealed diminishing returns to overlay strategies beyond approximately 15–20 % of portfolio value, with the high derivative allocation scenario (20 %) achieving a Sharpe ratio of 1.92 compared to the baseline 1.85, representing only a modest 4 % improvement despite the 30 % increase in derivative exposure. The low derivative scenario (10 %) produced notably worse results with a Sharpe ratio of 1.77 and maximum drawdown of 15.8 %, suggesting that some minimum level of derivative overlay proves essential for effective tail risk management. The scenario eliminating derivative overlays entirely experienced the poorest performance with a Sharpe ratio of 1.58 and maximum drawdown of 17.5 %, comparable to the unoptimized benchmark and highlighting the substantial contribution of derivative strategies to overall risk-adjusted performance [39]. analyzed optimal derivative allocation levels and reported target ranges of 12–18 % for equity-dominated portfolios and 8–15 % for balanced portfolios, consistent with our baseline 15.3 % allocation. The comprehensive sensitivity analysis demonstrated that while the framework's performance exhibited some dependence on parameter specifications, the qualitative superiority versus benchmark approaches persisted across all reasonable parameter ranges. Even the worst-performing sensitivity scenario (no derivative overlays) achieved a Sharpe ratio of 1.58, more than double the benchmark's 0.73, suggesting that the fundamental optimization approach rather than precise parameter calibration drives the majority of performance enhancement. This robustness provides confidence that the framework can deliver consistent outperformance across varying market environments and implementation details, a critical consideration for practical institutional deployment.

4. Conclusion, policy recommendations, limitations, and future research directions

This study developed and evaluated an integrated financial risk management framework that unifies multi-objective portfolio optimization, coherent tail-risk measurement, copula-based dependency modeling, derivative overlay implementation, Monte Carlo simulation under stochastic volatility, structured stress testing, attribution analytics, and dynamic rebalancing with transaction-cost awareness. Empirical evidence from the 2019–2024 evaluation period shows that the proposed architecture materially improves institutional portfolio robustness relative to conventional benchmark approaches by simultaneously enhancing risk-adjusted performance and reducing left-tail losses. The optimized portfolio achieved higher and more stable risk-adjusted returns, as reflected in the Sharpe ratio and Information ratio, while delivering a meaningful reduction in maximum drawdown and consistent improvements in VaR/CVaR across multiple horizons and confidence levels. The results further indicate that the framework's advantage is not driven by a single component, but rather by the interaction of (i) tail-risk-aware optimization that constrains extreme losses, (ii) adaptive dependence modeling that captures regime-dependent correlation and tail co-movement, and (iii) systematic derivative overlays that reshape the portfolio payoff distribution by limiting downside convexity costs during crises and harvesting volatility premia selectively. Importantly, the framework remained resilient under inflation shocks, credit crises, liquidity crunches, and currency stress scenarios, highlighting its practical relevance for institutional asset managers operating in environments where traditional diversification can fail, particularly when equity–bond correlations temporarily shift toward positive regimes.

From a policy and regulatory perspective, the findings support several actionable recommendations aimed at strengthening systemic resilience and improving governance standards in institutional portfolio management. First, regulatory risk frameworks should

move beyond single-point VaR reporting and adopt coherent tail-risk measures—particularly Expected Shortfall (CVaR)—as a primary supervisory metric across institutions, since CVaR better captures the severity of extreme losses and provides incentives aligned with diversification under stress. Second, policymakers and supervisors should encourage standardized stress testing protocols that combine macro narrative scenarios (inflation regime shifts, credit spread shocks, currency crises, liquidity freezes) with transparent model governance and disclosure requirements, because portfolio fragility often emerges from correlation breakdowns and liquidity dynamics that are not captured by normal-time volatility metrics. Third, institutional risk governance should explicitly mandate model risk management standards for dependence modeling, including periodic validation of correlation regime changes and tail dependence diagnostics, as the results demonstrate that static correlation assumptions can materially underestimate downside co-movement precisely when protection is most needed. Fourth, given the demonstrated benefits and potential pitfalls of derivatives, regulators and institutional boards should prioritize robust derivative governance frameworks that require clear risk limits on Greeks exposures, scenario-based margin stress assessments, and liquidity-aware hedging rules to reduce procyclical liquidation risk during volatility spikes. Fifth, at a market-structure level, continued support for transparent, liquid derivatives markets and robust central clearing infrastructures can improve the feasibility of low-cost hedging and reduce counterparty risk; however, this must be paired with prudent margining and liquidity buffers to avoid destabilizing fire-sales triggered by rapid collateral calls during systemic stress. Despite its contributions, the study has several limitations that should be considered when interpreting results and applying the framework in live investment settings. First, while the analysis incorporates DCC-GARCH correlations, copula dependence, and Heston stochastic volatility, all such models remain approximations of complex market dynamics and may fail under unprecedented structural breaks, regime shifts, or policy discontinuities that lie outside the calibration sample. Second, the reported performance metrics depend on assumptions about transaction costs, slippage, and derivative execution quality; real-world implementation—particularly at very large scale—may face additional market impact costs, execution constraints, and capacity limitations, especially in less liquid asset classes such as emerging markets, high yield credit, and certain alternatives. Third, the integration of alternative assets is inherently constrained by valuation smoothing, infrequent pricing, and illiquidity, which can bias volatility and tail-risk estimates downward and may understate true crisis-time co-movement; while the framework partially mitigates this through stress testing and dependence modeling, measurement error remains a persistent challenge. Fourth, the derivative overlay analysis assumes stable access to liquid options and futures markets and relies on model-based pricing and hedging relationships that can break down when implied volatility surfaces shift abruptly, liquidity evaporates, or bid–ask spreads widen during crisis periods. Fifth, the framework focuses primarily on market and credit risk dimensions; operational risks (model governance failures, data quality issues, cyber risks), legal constraints, and behavioral responses of market participants were not explicitly modeled, though these factors can materially affect realized outcomes for institutional portfolios.

Future research can extend the proposed framework in several directions to increase realism, scalability, and policy relevance. Methodologically, one promising avenue is to incorporate regime-switching or macro-state-dependent optimization, in which portfolio objectives, constraints, and hedging intensity adapt endogenously to inflation regimes, liquidity states, and monetary policy reaction functions rather than relying on fixed risk-aversion parameters. Additional enhancements could integrate higher-frequency microstructure-aware execution models to estimate implementation shortfall and endogenous market impact, enabling more accurate assessment of rebalancing frequency tradeoffs for large institutions. The dependence modeling layer can be advanced through time-varying copulas, vine copula structures, or nonparametric tail dependence estimators that better capture multivariate extreme co-movements across many asset classes simultaneously, particularly during cross-market deleveraging episodes. On the derivatives side, future work should evaluate robust hedging strategies under margin constraints and collateral dynamics, explicitly modeling funding liquidity and the interaction between hedging decisions and margin calls, as these channels often amplify losses during crises. From a data and validation perspective, the framework can be tested across longer historical windows spanning multiple inflation and rate regimes, and across different jurisdictions, to assess generalizability under varying market structures and regulatory environments. Finally, future research could incorporate climate and geopolitical risk factors—such as energy price shocks, transition risk, sanctions, and supply-chain disruptions—into stress testing narratives and factor decompositions, thereby aligning institutional risk management with emerging macro-financial vulnerabilities that increasingly drive tail events in global markets.

CRediT authorship contribution statement

Jiayang Wang: Conceptualization, Methodology, Formal analysis, Writing – original draft. **Zhuoyang Zhao:** Data curation, Investigation, Validation, Writing – review & editing. **Li Wang:** Resources, Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] F. Esmaeilion, M. Soltani, A. Luna-Triguero, Soft computing optimization of a renewable energy-integrated multigeneration system with liquid air energy storage, J. Energy Storage 127 (2025) 117140, https://doi.org/10.1016/j.est.2025.117140.
[2] A. Mahmoudan, P. Samadof, S. Hosseinzadeh, D.A. Garcia, A multigeneration cascade system using ground-source energy with cold recovery: 3E analyses and multi-objective optimization, Energy 233 (2021) 121185, https://doi.org/10.1016/j.energy.2021.121185.
[3] F. Esmaeilion, M. Soltani, Novel hybridization concept for efficient performance of a peak-shaving technology by adsorption and liquid air energy storage systems, Energy Convers. Manag. 343 (2025) 120183, https://doi.org/10.1016/j.enconman.2025.120183.
[4] H. Jafarizadeh, M. Soltani, W. Alfraidi, A comprehensive thermoeconomic assessment of liquid air and compressed air energy storage with solid/liquid/hybrid thermal energy storage (TES): addressing air and TES material storage cost impacts, Appl. Energy 388 (2025) 125615, https://doi.org/10.1016/j.apenergy.2025.125615.
[5] R.A. Badeeb, B. Wang, J. Zhao, Z. Khan, K.F. Uktamov, C. Zhang, Natural resources extraction and financial inclusion: Linear and non-linear effect of natural resources on financial sector, Resour. Policy 85 (2023), https://doi.org/10.1016/j.resourpol.2023.103826.
[6] D. Yang, Coping with disaster: the impact of hurricanes on international financial flows, 1970-2002, B E J. Econ. Anal. Pol. 8 (1) (2008), https://doi.org/10.2202/1935-1682.1903.
[7] L. Pan, Y. Wang, X. Sun, M. Sadiq, A.A. Dagestani, Natural resources: a determining factor of geopolitical risk in Russia? Revisiting conflict-based perspective, Resour. Policy 85 (2023), https://doi.org/10.1016/j.resourpol.2023.104033.
[8] E.F. Fama, K.R. French, Common risk factors in the returns on stocks and bonds, J. Financ. Econ. 33 (1) (1993) 3–56, https://doi.org/10.1016/0304-405X(93)90023-5.
[9] D.M. Franks, R. Davis, A.J. Bebbington, S.H. Ali, D. Kemp, M. Scurrah, Conflict translates environmental and social risk into business costs, Proc. Natl. Acad. Sci. U. S. A 111 (21) (2014) 7576–7581, https://doi.org/10.1073/PNAS.1405135111.
[10] M. Hodula, B. Škrabić Perić, P. Sorić, Economic uncertainty and non-bank financial intermediation: evidence from a European panel, Finance Res. Lett. 53 (2023) 103675, https://doi.org/10.1016/j.frl.2023.103675.
[11] Y. Hu, Y. Hou, Greg, L. Oxley, S. Corbet, Does blockchain patent-development influence bitcoin risk? J. Int. Financ. Mark. Inst. Money 70 (2021) 101263 https://doi.org/10.1016/j.intfin.2020.101263.
[12] D.W. Diamond, Financial intermediation and delegated monitoring, Rev. Econ. Stud. 51 (3) (1984) 393–414, https://doi.org/10.2307/2297430.
[13] V. Singh, E.D. Roca, China's geopolitical risk and international financial markets: evidence from Canada, Appl. Econ. 54 (34) (2022) 3953–3971, https://doi.org/10.1080/00036846.2021.2019185.
[14] Y. Ji, L. Shi, S. Zhang, Digital finance and corporate bankruptcy risk: evidence from China, Pac. Basin Finance J. 72 (2022), https://doi.org/10.1016/j.pacfin.2022.101731.
[15] Y. Cai, M. Umair, N.A. Adam, R. Zhang, S. Mirzaliev, C. Chang, Marketization as a catalyst: understanding the impact of digital economies on green innovation in mainland China, Manag. Decis. Econ. (2025), https://doi.org/10.1002/mde.4509 *n/a*(n/a).
[16] Y. Song, B. Chen, X.Y. Wang, P.P. Wang, Defending global oil price security: based on the perspective of uncertainty risk, Energy Strategy Rev. 41 (2022), https://doi.org/10.1016/J.ESR.2022.100858.
[17] H. Yuan, L. Zhao, M. Umair, Crude oil security in a turbulent world: china's geopolitical dilemmas and opportunities, Extr. Ind. Soc. 16 (2023) 101334, https://doi.org/10.1016/j.exis.2023.101334.
[18] X. Zhang, Y. Tian, C.C. Lee, Enforcement actions and systemic risk, Emerg. Mark. Rev. 59 (2024), https://doi.org/10.1016/j.ememar.2024.101115.
[19] Q. Lu, M. Umair, Z. Qin, M. Ullah, Exploring the nexus of oil price shocks: impacts on financial dynamics and carbon emissions in the crude oil industry, Energy 312 (2024) 133415, https://doi.org/10.1016/j.energy.2024.133415.
[20] S. Qu, J. Wang, Y. Li, K. Wang, How does risk-taking affect the green technology innovation of high-tech enterprises in China: the moderating role of financial mismatch, Environ. Sci. Pollut. Control Ser. 30 (9) (2023) 23747–23763, https://doi.org/10.1007/S11356-022-23820-0.
[21] X. Zheng, M. Imran, M. Umair, C. Lin, Y. Dong, Modern panelizing and optimization techniques for renewable energy projects; perspectives on how CO2 emissions impact the circular economy, Energy 323 (2025) 135881, https://doi.org/10.1016/j.energy.2025.135881.
[22] D. Kim, C. Li, X. Wang, Risk-taking and performance of government bond mutual funds, Int. Rev. Financ. Anal. 76 (2021), https://doi.org/10.1016/J.IRFA.2021.101780.
[23] X. Cui, M. Umair, G. Ibragimove Gayratovich, A. Dilanchiev, DO remittances mitigate poverty? AN empirical evidence from 15 selected asian economies, Singapore Econ. Rev. 68 (4) (2023) 1447–1468, https://doi.org/10.1142/S0217590823440034.
[24] Z. Bao, W. Ji, S. Abdullayeva, V. Teymurova, S. Ali, Investor attention, ESG strength, and green innovation efficiency: a two-stage network DEA approach for Chinese listed manufacturers, Int. Rev. Econ. Finance (2025) 104782, https://doi.org/10.1016/j.iref.2025.104782.
[25] Q.T. Tran, Economic policy uncertainty and corporate risk-taking: international evidence, J. Multinatl. Financ. Manag. (2019) 52–53, https://doi.org/10.1016/j.mulfin.2019.100605.
[26] Z. Ahmed, M. Ahmad, R. Alvarado, A. Sinha, M.I. Shah, S. Abbas, Towards environmental sustainability: do financial risk and external conflicts matter? J. Clean. Prod. 371 (2022) https://doi.org/10.1016/J.JCLEPRO.2022.133721.
[27] D.T.T. Nguyen, I. Diaz-Rainey, H. Roberts, M. Le, Loans from my neighbours: east Asian commercial banks, financial integration, and bank default risk, Int. Rev. Financ. Anal. 74 (2021), https://doi.org/10.1016/J.IRFA.2021.101659.
[28] C.G. Ntim, S. Lindop, D.A. Thomas, Corporate governance and risk reporting in South Africa: a study of corporate risk disclosures in the pre- and post-2007/2008 global financial crisis periods, Int. Rev. Financ. Anal. 30 (2013) 363–383, https://doi.org/10.1016/j.irfa.2013.07.001.
[29] J. Thomä, H. Chenet, Transition risks and market failure: a theoretical discourse on why financial models and economic agents may misprice risk related to the transition to a low-carbon economy, J. Sustain. Financ. Invest. 7 (1) (2017) 82–98, https://doi.org/10.1080/20430795.2016.1204847.
[30] D.A. Kesto, J. Ravi, Capital budgeting practices in financial institutions (FIs): an empirical study in the case of Ethiopia, Global Disclosure of Economics and Business 6 (1) (2017) 29–40, https://doi.org/10.18034/GDEB.V6I1.114.
[31] R.E. Krainer, Financial contracting as behavior towards risk: the corporate finance of business cycles, J. Financ. Stabil. 65 (2023) 101104, https://doi.org/10.1016/J.JFS.2023.101104.
[32] B. Wang, C. Yan, N. Iqbal, Z. Fareed, A. Arslan, Impact of human capital and financial globalization on environmental degradation in OBOR countries: critical role of national cultural orientations, Environ. Sci. Pollut. Control Ser. 29 (25) (2022) 37327–37343, https://doi.org/10.1007/S11356-022-18556-W.
[33] Y. Shahab, C.G. Ntim, F. Ullah, C. Yugang, Z. Ye, CEO power and stock price crash risk in China: do female directors' critical mass and ownership structure matter? Int. Rev. Financ. Anal. 68 (2020) 101457 https://doi.org/10.1016/j.irfa.2020.101457.
[34] H.T. Pham, C. Gan, B. Hu, Causality between financial development and foreign direct investment in Asian developing countries, J. Risk Financ. Manag. 15 (5) (2022), https://doi.org/10.3390/JRFM15050195.
[35] S.T. Ergasheva, A.A. Tillyakhodjaev, Y.K. Karrieva, E.G. Popkova, Z.V. Gornostaeva, Business risks in COVID-19 crisis dataset modeling: regulatory vs. marketing tools of risk management, Risks 11 (11) (2023), https://doi.org/10.3390/RISKS11110190.
[36] L. Xu, W. Chen, S. Wang, B.S. Mohammed, R. Lakshmana Kumar, Analysis on risk awareness model and economic growth of finance industry, Ann. Oper. Res. (2022) 1–22, https://doi.org/10.1007/s10479-021-04516-z.
[37] C. Luo, Y. Qu, Y. Su, L. Dong, Risk spillover from international crude oil markets to China's financial markets: evidence from extreme events and U.S. monetary policy, N. Am. J. Econ. Finance 70 (2024) 102041, https://doi.org/10.1016/j.najef.2023.102041.
[38] M. Spiegel, H. Zhang, Mutual fund risk and market share-adjusted fund flows, J. Financ. Econ. 108 (2) (2013) 506–528, https://doi.org/10.1016/J.JFINECO.2012.05.018.
[39] A. Hadian, C. Adaoglu, The effects of financial and operational hedging on company value: the case of Malaysian multinationals, J. Asian Econ. 70 (2020).